INDUSTRIAL ELECTRONICS N5 (8080175)

PROGRAMME(S):ELECTRICAL ENGINEERING ALL GROUPS

LECTURER(S): MR HS MBELE

CHAPTER 05

HANDLING CMOS

OUTLINE

1. Learning outcomes
2. Introduction
3. Content
    a) Concept
    b) Examples
4. Activity

1.OUTCOMES

- Be able to describe with the aid of a neat labelled diagram the protection of a CMOS IC
- Explain the methods of handling CMOS

INTEGRATED CIRCUITS(I.C)

- This is a small electronic device made of silicon on which components such as resistors, capacitors, transistors are mounted which work together to perform specific tasks.

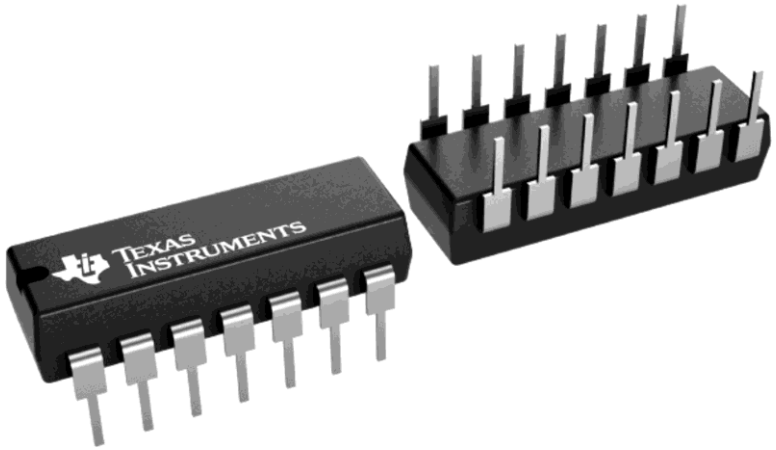
- **IC CATEGORIES**

- Small scale integration(SSI) – Ten to hundreds of transistors per chip.
- Medium scale integration (MSI) – Hundreds to hundred thousand per chip
- Very large scale integration (VLST) – Up to 1 million per chip
- Ultra-large scale (ULSI) – Billions per chip

- **APPLICATIONS INCLUDE**

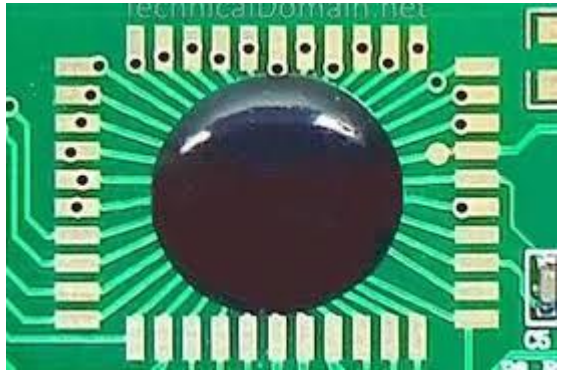
- Computing, medical devices, Aerospace, defense, internet of things, automotive industry etc

TYPICAL EXAMPLES OF IC PACKAGES









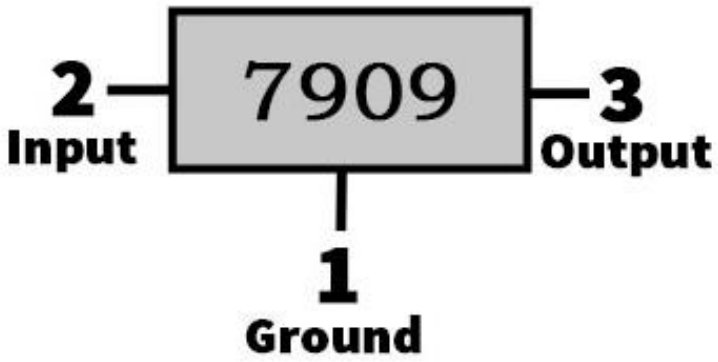


INTEGRATED CIRCUIT VOLTAGE REGULATORS

- This is small THREE terminal electronic device designed to automatically maintain a consistent voltage level.

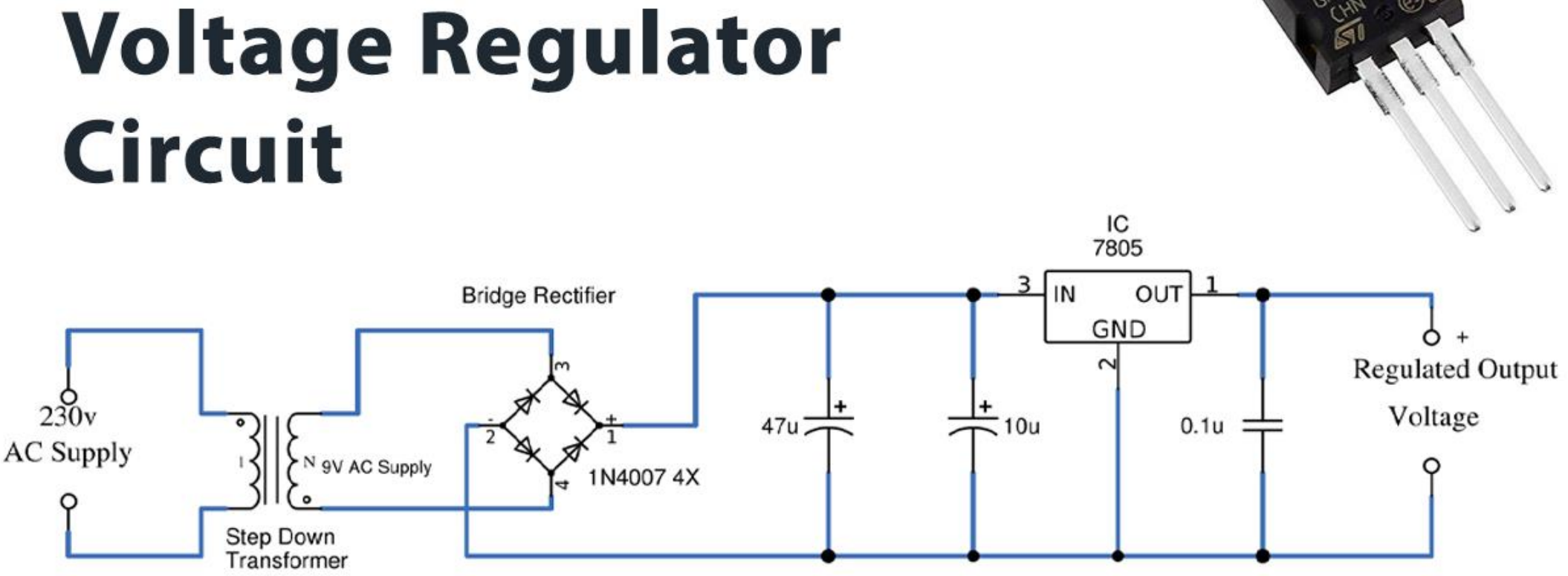
LM7909 Pinout



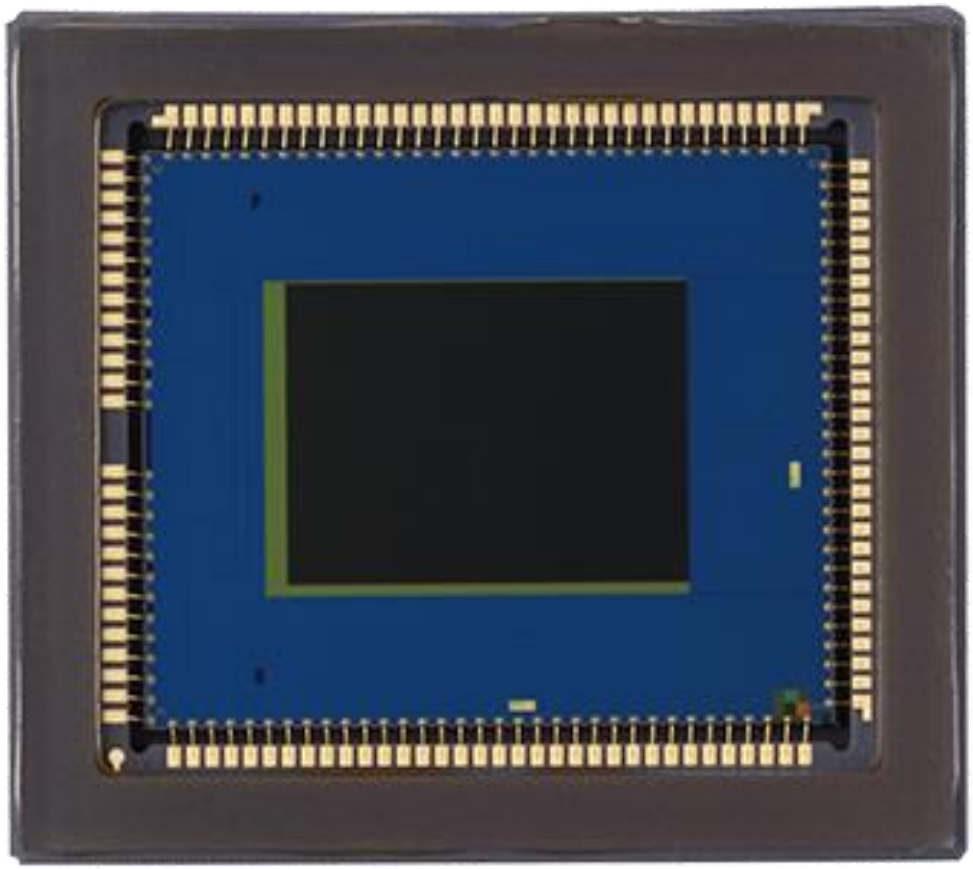


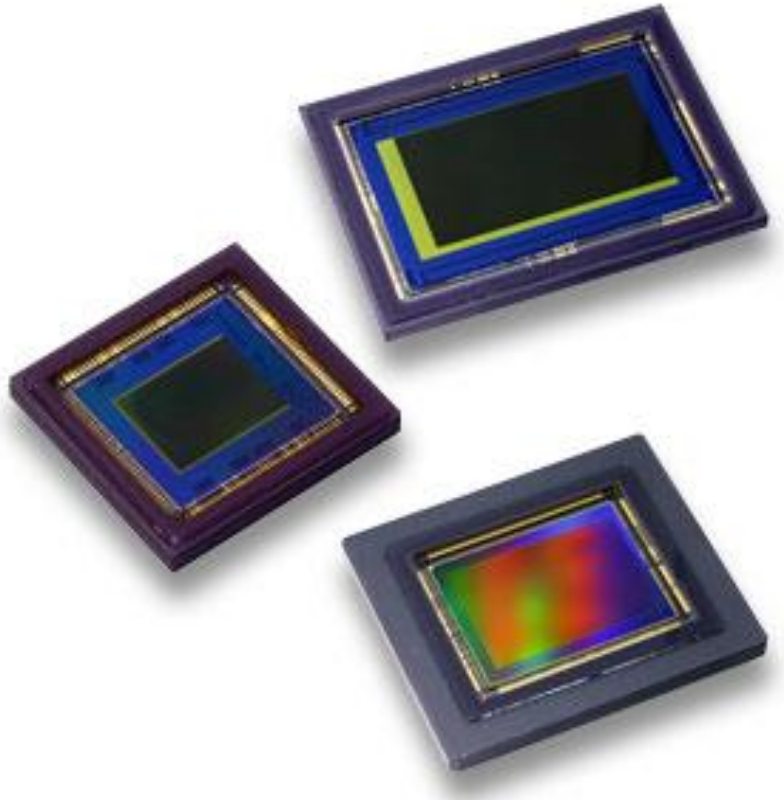
IC 7812 REGULATOR IN A BRIDGE RECTIFIER





C-MOS DEVICES



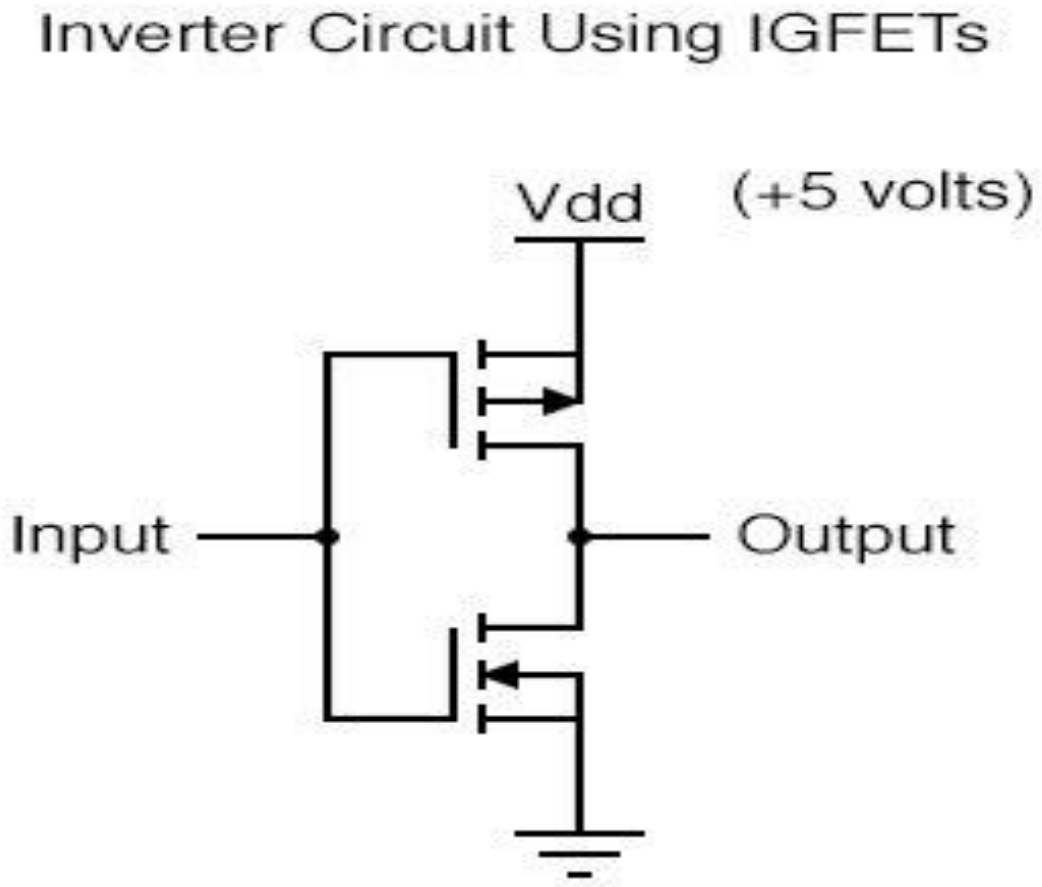


C-MOS DEVICES

- **C-MOS** – Complementary metal oxide semiconductor
- The term complimentary refers to the fact that the typical design style of C-MOS uses complementary and symmetrical pairs of **P-type** and **N-type** metal oxide semiconductor field effect transistors for logic functions.
- The term **Metal oxide semiconductor** is reference to the physical structure of certain field effect transistors, having metal gate electrode placed on top of an oxide insulator which in turn is on top of a semiconductor.
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- Used in micro processors, micro controllers etc
- Used also in other analog circuits such as image sensors and data convertors.
- Used also for battery operated applications

3. TWO IMPORTANT CHARACTERISTICS OF C-MOS

- Noise immunity
- Power consumption is low



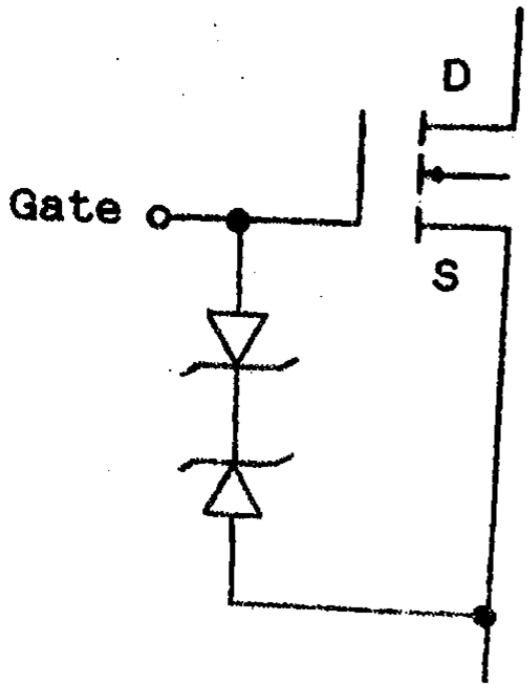
3. INPUT PROTECTION

- Due to the fact that these semiconductor devices are highly affected by high electrical fields in the gate oxide regions, the input needs to be well protected against these static fields.

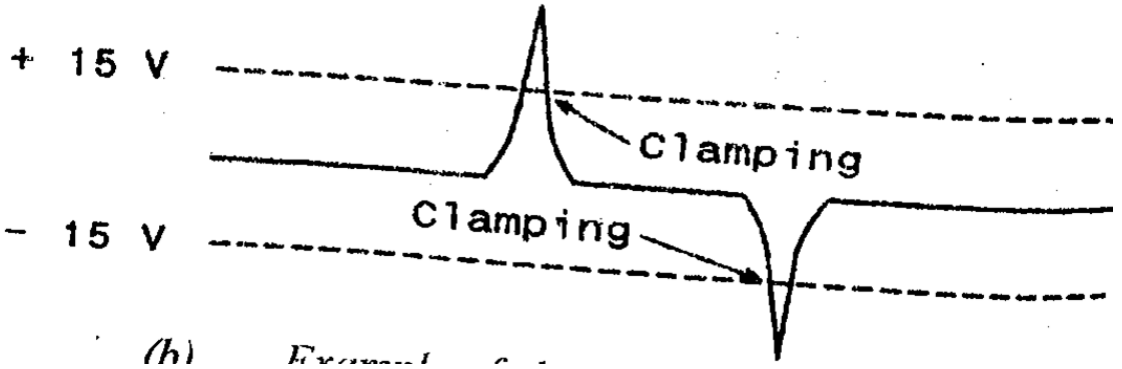
Protection method

- C-MOS are affected by high electrical fields in the gate regions and static pick up.
- Diodes are used to protect C-MOS from high electrical fields.
- Diode are connected in reverse bias configuration between the input and output and are called Voltage limiters.

3. INPUT PROTECTION



(a) Zener-diode gate protection



2 (b) Example of

3. PROCEDURES TO BE FOLLOWED WHEN HANDLING C-MOS

- Store unutilized units in a conductive sponge
- Use earthed point soldering irons as soon as you solder the joints
- Use specially earthed wristbands
- Connect all test equipment to earth
- Switch off power before you work on the circuit
- All unused inputs must be connected to V_{DD} or V_{ss}

4. NOTES